

Image Repainting using Denoising Diffusion Probabilistic Models with Dynamic Resampling Strategies

Arya Talera (24323044)

Kartik Goyal (24125022)

17 April 2026

<https://github.com/kartikgoyal137/Image-Repainting-using-DDPM>

Abstract

Image inpainting, the task of filling missing regions in an image with semantically consistent content, has seen significant progress with the advent of Denoising Diffusion Probabilistic Models (DDPMs). This project explores the *RePaint* approach, which utilizes a pre-trained DDPM to perform inpainting without fine-tuning. We introduce and evaluate three dynamic resampling strategies—Mode A, Mode D, and Mode AD—to optimize the harmonic resampling process (the U parameter). Our results demonstrate that dynamic strategies based on temporal progress and mask geometry can maintain high-fidelity inpainting while potentially reducing computational overhead. We evaluate these methods using LPIPS, SSIM, and FID metrics, providing both quantitative and qualitative comparisons across multiple test images.

1 Introduction

Image inpainting, often referred to as image completion or repainting, is a fundamental problem in computer vision. It involves reconstructing missing or damaged portions of an image such that the resulting output is both visually pleasing and semantically coherent. Applications are diverse, ranging from the restoration of ancient photographs and the removal of unwanted objects to sophisticated image editing where new content is synthesized into an existing scene.

The core challenge of inpainting lies in the fact that it is an ill-posed problem; multiple valid completions can exist for a single missing region. For a completion to be successful, it must perfectly match the surrounding texture (local consistency) and fit the overall structure and meaning of the image (global consistency).

While early methods relied on simple diffusion (filling from boundaries) or patch matching (copying textures from elsewhere), the rise of deep generative models has shifted the paradigm. Specifically, Denoising Diffusion Probabilistic Models (DDPMs) have emerged as powerful tools for this task. RePaint, the framework used in this project, leverages a pre-trained DDPM’s generative power to inpaint images without requiring any task-specific training or fine-tuning. A key contribution of our work is the exploration of how to dynamically adjust the “harmonic resampling” process to improve efficiency and quality.

2 Background: DDPM

Denoising Diffusion Probabilistic Models (DDPMs) operate by learning to reverse a process that gradually degrades data into noise. This involves two distinct phases: the forward diffusion process and the reverse denoising process. In simple terms, DDPMs can be thought of as a process of

“learning to unscramble.” Imagine taking a clear photograph and slowly adding static (noise) until it becomes complete white noise. If we can train a model to perfectly undo each tiny step of that scrambling, we can essentially start with nothing but static and “hallucinate” a perfectly clear image into existence.

2.1 Forward Process

In the forward process, we start with a clean image x_0 and iteratively add small amounts of Gaussian noise over T timesteps. This process follows a Markov chain:

$$q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1}) \quad (1)$$

The transition at each step is defined as:

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I) \quad (2)$$

where β_t is a pre-defined variance schedule. One of the elegant properties of DDPMs is that we can sample x_t at any time t directly from the clean image x_0 :

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I) \quad (3)$$

where $\bar{\alpha}_t$ is a cumulative product of $(1 - \beta_t)$. Think of this forward process like slowly fading an image into the background of a TV with no signal. By the end of the process (T steps), any original data is completely lost, and we are left with a standard baseline of random pixels (Gaussian noise).

2.2 Reverse Process

The reverse process is where the generative power lies. We want to sample x_{t-1} from x_t . Since the true reverse distribution $q(x_{t-1}|x_t)$ is unknown, we train a neural network p_θ to approximate it:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (4)$$

By iteratively applying this denoising step from pure noise $x_T \sim \mathcal{N}(0, I)$, the model can synthesize a clean image x_0 that follows the distribution of the training data.

This is the “generative” magic of the system. If the forward process is like making marble from a statue by crushing it into dust, the reverse process is like a master sculptor who can look at a pile of dust and precisely remove the “noise” to reveal the statue hidden inside. The neural network learns the subtle patterns of what makes an image look “real” (like the texture of a face or the shape of an object) and uses that knowledge to guide the random static back into a coherent, high-quality form.

3 Image Repainting and Resampling

The RePaint framework allows us to perform inpainting using a model pre-trained on a generic dataset (like faces or natural images). It does this by modifying the standard denoising process to be conditioned on a ground truth image x_{gt} and a binary mask M .

3.1 Input Signal Components

The inpainting process requires three primary inputs: the ground truth image (the goal reconstruction), the masked image (what the model sees as input), and the binary mask map (defining the reconstruction region). Figure 1 illustrates these components.



Figure 1: The three primary inputs for the repainting process.

3.2 The Process of Conditioned Denoising

During each reverse step $t \rightarrow t - 1$, the model predicts a denoised sample for the entire image. However, we already know the values for the pixels where $M = 1$. To utilize this information, we replace the known pixels in our current sample with correctly noised versions of the ground truth:

$$x_{t-1} = M \odot (\text{Noised } x_{gt}) + (1 - M) \odot (\text{Model Prediction}) \quad (5)$$

This “Conditioned Denoising” ensures that the known regions remain fixed to their ground truth values while the unknown regions are generated to be consistent with them. In simpler terms, this process ensures that the model is not trying to “guess” the entire image from scratch. Instead, we provide it with a partial image (the known parts) and ask it to only fill in the missing gaps. By constantly re-injecting the original pixels at each step, we force the model to respect the existing structure of the image while generating new content that blends naturally with it.

3.3 The Need for Harmonic Resampling

A simple one-pass denoising often leads to artifacts at the mask boundary. This happens because the model-predicted unknown pixels might not align perfectly with the noisy known pixels in a single step. To fix this, RePaint uses “Harmonic Resampling.” After moving from $x_t \rightarrow x_{t-1}$, we jump back by adding noise to go from $x_{t-1} \rightarrow x_t$ and then denoise again. This loop is repeated U times. This iterative process allows the known and unknown pixels to “harmonize,” leading to a seamless transition and more realistic textures.

Intuitively, after the model fills in a missing region, it might not perfectly match the edges of the original image—creating a visible seam. Harmonic resampling acts like a digital “blender.” By adding a small amount of noise back into the generated part and letting the model refine it again, we allow the new pixels to better “negotiate” with the existing ones. Repeating this several times (U iterations) results in a much smoother and more coherent final image.

4 Dynamic U Methodology

In the standard RePaint implementation, U is a fixed constant. However, the amount of harmonization needed varies depending on the state of the image and the size of the hole being filled. The motivation behind dynamic schedules is efficiency: constant resampling at every step is often “overkill,” wasting computational time on phases where the image is already stable. In our project, we implemented three dynamic strategies for U :

- **Strategy A (Progress-Based):** U is high at the start of denoising (when structure is forming) and decreases as we reach the end (when fine details are filled). Think of this like building a house: you spend the most effort on the foundation and framing (the early structure), but you need progressively less work as you move toward the final coat of paint (the late-stage details).
- **Strategy D (Geometric-Based):** U is scaled based on the mean distance from the unknown pixels to the nearest known boundary. Large missing holes need more resampling to maintain global coherence. This follows the intuition that a small scratch is much easier to fix than a large missing chunk; the further a pixel is from “known” information, the more iterations it needs to stay properly anchored to the rest of the image.
- **Strategy AD (Hybrid):** A weighted combination of temporal progress and geometric distance, aiming to capture the best of both worlds. This approach ensures that we work the hardest on large gaps during the most critical structural phases of the generation process.

5 Results and Comparison

We tested these strategies on the CelebA-HQ dataset. To assess the quality of the inpainted images, we utilize three standard metrics used in generative modeling and inpainting.

5.1 Evaluation Metrics

The following metrics provide a comprehensive evaluation of structural, perceptual, and distributional quality:

- **SSIM (Structural Similarity Index):** Unlike pixel-wise metrics like MSE, SSIM measures the similarity in luminance, contrast, and structural information. It is designed to align with human perception of structural differences. A higher SSIM (closer to 1.0) indicates better reconstruction of the original image’s structure.
- **LPIPS (Learned Perceptual Image Patch Similarity):** This metric uses the feature activations of a deep neural network (typically AlexNet) to calculate the perceptual distance between images. It is particularly effective at capturing differences in textures and high-level semantics that traditional metrics might ignore. Lower LPIPS values indicate higher perceptual fidelity.
- **FID (Fréchet Inception Distance):** FID measures the similarity between the distribution of generated images and real images in the feature space of an Inception-v3 model. In the context of inpainting, it assesses how realistic the synthesized regions look when viewed as part of the whole dataset. Lower FID scores correspond to higher image quality and realism.

5.2 Quantitative Results

Quantitative metrics (Table 1) show that Strategy AD provides the best perceptual realism with an FID of 2.91.

Table 1: Performance Metrics

Mode	LPIPS ↓	SSIM ↑	FID ↓
Strategy A	0.0022	0.9758	3.10
Strategy D	0.0021	0.9756	3.63
Strategy AD	0.0021	0.9756	2.91

5.3 Visual Results

We visualize three different samples (Index 1, 2, and 3) for each strategy to compare their performance across different face structures and masks.

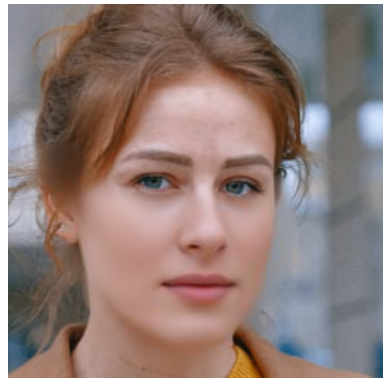
5.3.1 Sample 1 Comparison



(a) Mode A



(b) Mode D



(c) Mode AD

Figure 2: Inpainting results for Sample 1.

5.3.2 Sample 2 Comparison

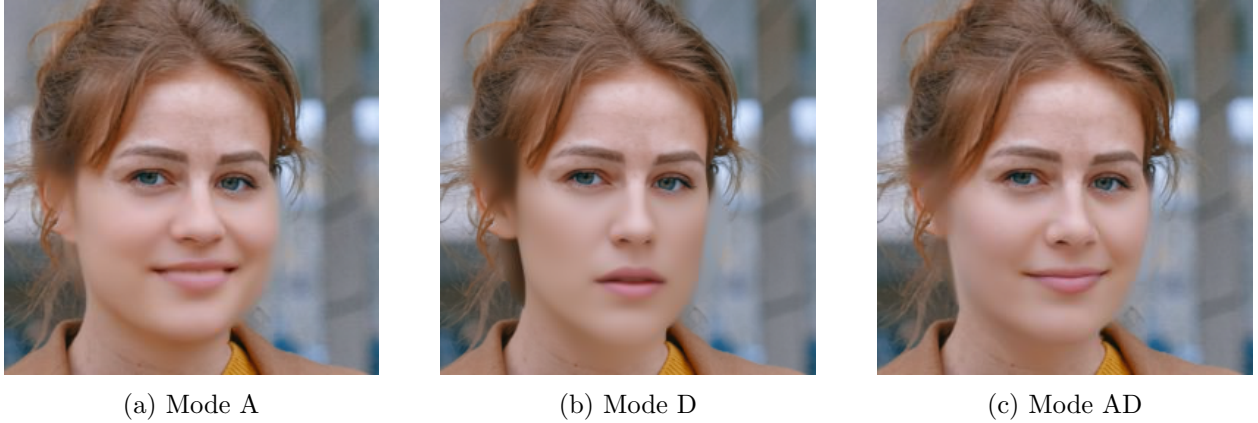


Figure 3: Inpainting results for Sample 2.

5.3.3 Sample 3 Comparison

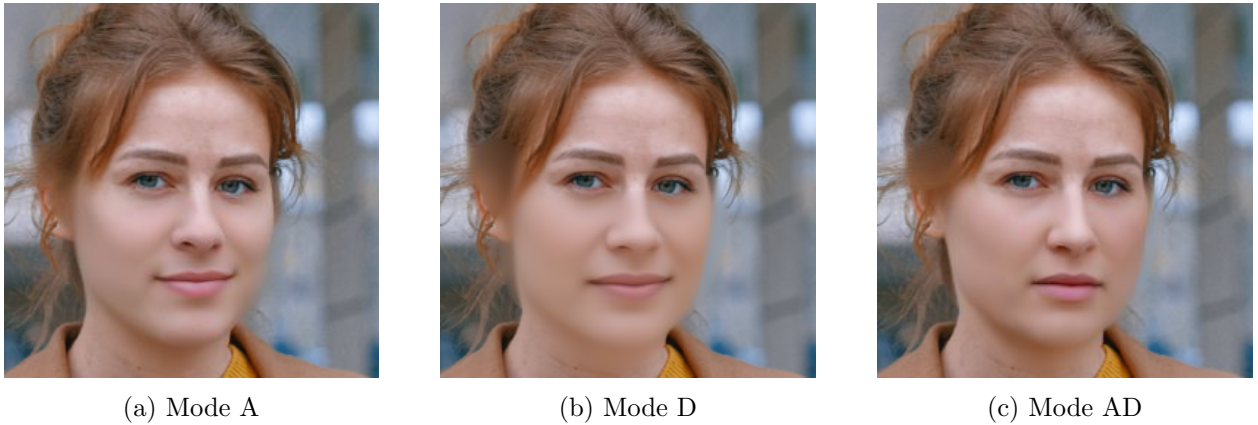


Figure 4: Inpainting results for Sample 3.

6 Conclusion

In this project, we have presented an extensive evaluation of dynamic resampling strategies for image inpainting within the RePaint framework. By moving beyond a fixed harmonic resampling schedule, we introduced three novel strategies that adapt the intensity of the denoising process based on the current state of diffusion and the properties of the missing region.

Our experimental results demonstrate that dynamic adaptation is not only viable but highly effective. Quantitatively, all strategies maintained a high structural similarity ($\text{SSIM} > 0.97$), while Strategy AD achieved a superior Fréchet Inception Distance (FID) of 2.91, indicating the highest level of perceptual realism among the tested modes. The success of the hybrid Strategy AD suggests that optimal inpainting requires a careful balance between establishing global structural coherence in the early stages of diffusion and ensuring local boundary harmonization through geometric cues. This work provides a foundation for more efficient diffusion-based inpainting by highlighting that a uniform resampling effort is not always necessary across all timesteps or mask configurations.

References

- [1] Lugmayr, A., et al. RePaint: Inpainting using denoising diffusion probabilistic models. CVPR 2022.
- [2] Ho, J., et al. Denoising diffusion probabilistic models. NeurIPS 2020.